



OBJECT ORIENTED SOFTWARE ENGINEERING

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Software Testing

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- Testing is the process of **exercising a program** with a specific intent of **finding errors** prior to delivery to the end user.
- **Software Testing** is a method to check whether the actual software product matches expected requirements and to ensure that software product is Defect free

- **Software testing** is a process used to identify the **correctness, completeness** and **quality** of the developed software.
- It also helps to **identify error, gaps** or **missing requirements**.
- The process include **verification** and **validation** of built software.

Verification and Validation

- During & after the implementation process, the program being developed must be checked to ensure that it **meets its specification** and **delivers the functionality** expected by people paying for software

– ***Verification:***

"Are we building the product right?"

– ***Validation:***

"Are we building the right product?"

- **Verification** (are the algorithms coded correctly?)
 - The set of activities that ensure that software **correctly implements a specific function.**
 - means Software confirms to its specification

- ***Verification*** is a static analysis technique.
- In this technique, **testing is done without executing the code.**
- Examples include – **Reviews, Inspection, and walkthroughs.**

- **Validation** (Does it meet user requirements?)
 - The set of activities that ensure that the software **that has been built** is traceable to **customer requirements**.
 - Means the software should do what the user really require

- ***Validation*** is a dynamic analysis technique **where testing is done by executing the code.**
- Examples include **functional** and **non-functional** testing techniques.

Verification

Validation

- | | | |
|---|--|---|
| 1 | Are you building it Right? | Have you build the right thing? |
| 2 | checks whether the software confirms a specification | checks whether the software meets the requirements and expectations. |
| 3 | Done by developer | Done by Tester |
| 4 | does not involve code execution | involves code execution. |
| 5 | Methods involves Review, Inspection, Unit Testing and Integration testing | Involves System Testing |
| 6 | Static and dynamic activities | Only Dynamic |



THANK YOU

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